BAYESIAN MODELING OF COMPLEXITY

Probability Foundations & Bayes' Law

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Agenda

1 Recap & Today

2 Probability Theory

- Random Variables
- PDF and CDF
- Joint Distributions
- Operations on Random Variables

3 Bayes' Law

- Conditional Probability
- Bayes' Law

4 Wrap-Up & Next

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Where We've Been: Sessions 1-3

What we've accomplished:

- ✓ Built ABMs of complex systems
- ✓ Implemented Schelling model
- ✓ Analyzed emergent patterns
- ✓ Conducted parameter sweeps

Open questions:

1. How do we **learn** model parameters from real data?
2. How do we **quantify uncertainty** in parameter estimates?
3. How do agents **update beliefs** based on observations?

The Answer: Bayesian Methods

Bayesian statistics provides a principled framework for:

- Parameter estimation with uncertainty quantification
- Model calibration and selection
- Adaptive agent behavior (agents as Bayesian reasoners)

Today's Learning Goals

By the end of this session, you will be able to:

1. **Work with probability distributions:**
   - Define random variables, PDF, CDF
   - Calculate joint and conditional probabilities
2. **Apply Bayes' Law:**
   - Derive Bayes' Law from conditional probability
   - Solve classic Bayesian problems (medical testing, etc.)
   - Interpret posterior probabilities
3. **Code in Python:**
   - Implement discrete Bayes updates
   - Visualize prior, likelihood, posterior

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The Big Picture: Statistics vs. Probability

Probability & **Statistics** provide a framework for quantifying uncertainty and reasoning about the likelihood of outcomes.

Statistics

Data → Model

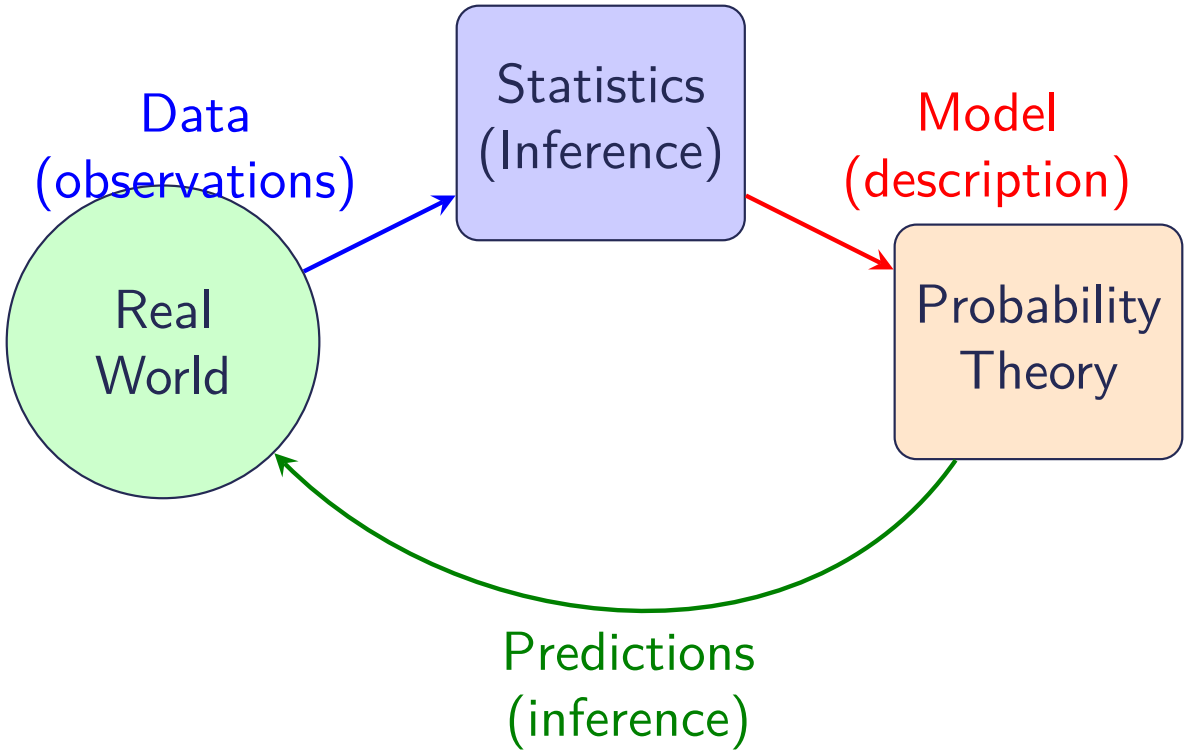
- Making sense of observations
- Descriptive: mean, variance, histograms
- Inferential: hypothesis testing, regression
- *Learn from data*

Probability Theory

Model → Predictions

- Quantifying uncertainty about events
- Rules for combining variables
- Forward predictions
- *Reason about uncertainty*

Real-World, Statistics, & Probability



What is Randomness?

Randomness: Lack of determinism in the outcome of an experiment

Deterministic:

- $y = 2x + 3$
- Given $x = 5 \xrightarrow{Always} y = 13$
- No uncertainty

Stochastic (Random):

- Flip a coin
- Multiple possible outcomes
- Can assign probabilities: $P(H) = 0.5$, $P(T) = 0.5$

Recall

Schelling model has stochastic elements:

- Initial agent placement (uniform random)
- Movement order (random shuffling)
- Destination choice (random empty cell)

These create *uncertainty* in the system

Random Variables (RVs)

Definition

A **Random Variable** X represents an experiment with multiple possible outcomes, each with an associated probability.

Notation:

- X: The random variable (capital letter)
- x: A specific value/outcome called **event**
- $P(X = x)$: Probability that X takes value x

Discrete RV Examples:

- Coin flip: $X \in \{H, T\}$
- Die roll: $X \in \{1, 2, 3, 4, 5, 6\}$
- Number of segregated neighborhoods in a city

Continuous RV Examples:

- Temperature: $X \in \mathbb{R}$
- Agent tolerance: $X \in [0, 1]$
- Final segregation index

Example: Coin Flip

Experiment: Flip a potentially unfair coin

Random Variable: X represents the outcome

Sample Space: $\Omega = \{H, T\}$ (all possible outcomes)

Probabilities:

$$P(X = H) = 0.65$$
$$P(X = T) = 0.35$$

Required properties:

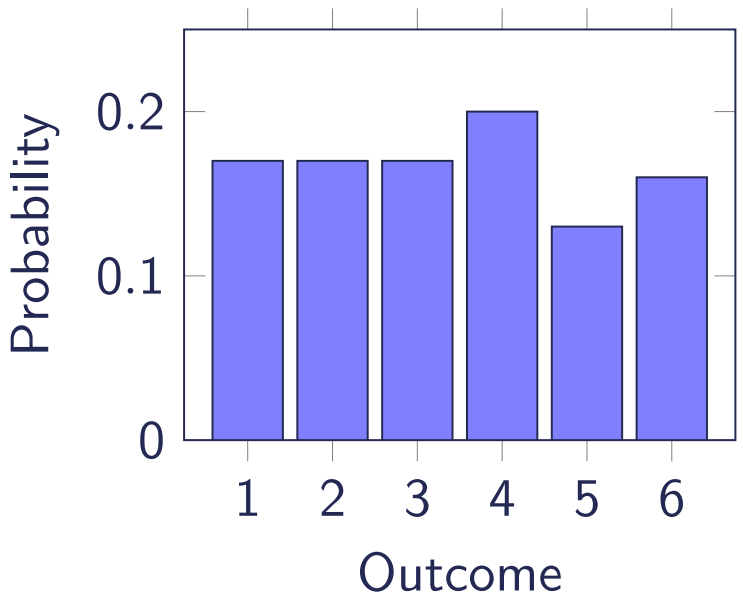
1. Non-negativity: $P(X = x) \geq 0$ for all x
2. Normalization: $\sum_x P(X = x) = 1$

Check: $0.65 + 0.35 = 1.0$

Example: Die Roll

Experiment: Roll a potentially biased die

$$P(X = 1) = 0.17$$
$$P(X = 2) = 0.17$$
$$P(X = 3) = 0.17$$
$$P(X = 4) = 0.20$$
$$P(X = 5) = 0.13$$
$$P(X = 6) = 0.16$$



Sum: 1.00

Probability Distribution Function (PDF)

Definition

The **PDF** $f_X(x)$ maps each possible outcome x to its probability $P(X = x)$.

For discrete RVs: Called **Probability Mass Function (PMF)**

For continuous RVs: $P(a \leq X \leq b) = \int_a^b f_X(x)dx$

Example: Unfair coin

$$f_X(x) = \begin{cases} 0.65 & \text{if } x = H \\ 0.35 & \text{if } x = T \end{cases}$$

Example: Biased die

$$f_X(x) = \begin{cases} 0.17 & \text{if } x \in \{1, 2, 3\} \\ 0.20 & \text{if } x = 4 \\ 0.13 & \text{if } x = 5 \\ 0.16 & \text{if } x = 6 \end{cases}$$

Cumulative Distribution Function (CDF)

Definition

The **CDF** $F_X(x)$ gives the probability that X takes a value less than or equal to x:

$$F_X(x) = P(X \leq x)$$

Relationship to PDF:

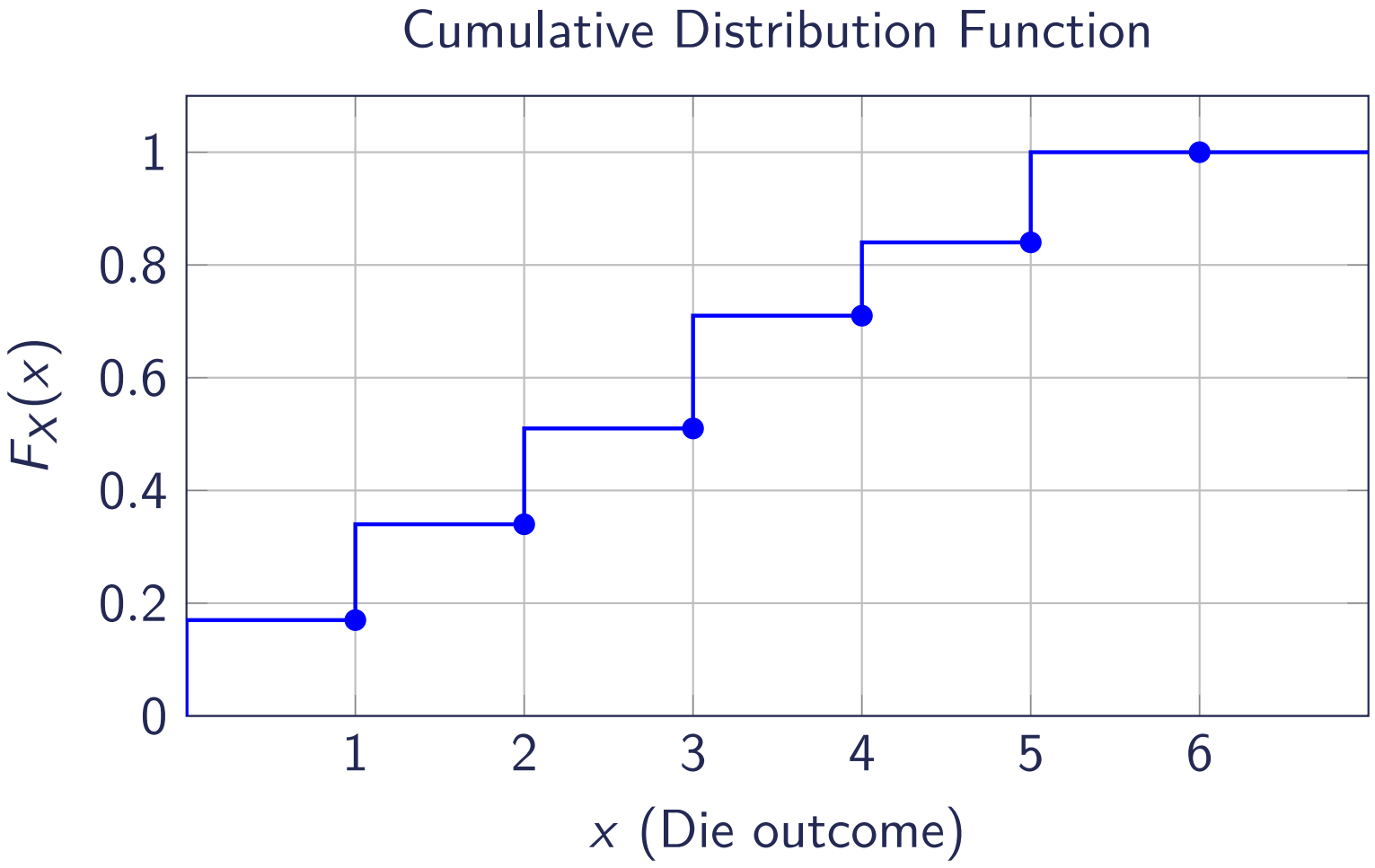
Discrete: $F_X(x) = \sum_{x_i \leq x} f_X(x_i)$

Continuous: $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$

Example: Die roll CDF

$$F_X(x) = \begin{cases} 0.17 & \text{if } x = 1 \\ 0.34 & \text{if } x = 2 \\ 0.51 & \text{if } x = 3 \\ 0.71 & \text{if } x = 4 \\ 0.84 & \text{if } x = 5 \\ 1.00 & \text{if } x = 6 \end{cases}$$

CDF Visualization



Question: What does $F_X^{-1}(0.25)$ represent?

Important Continuous Distribution: Gaussian (Normal)

Definition:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

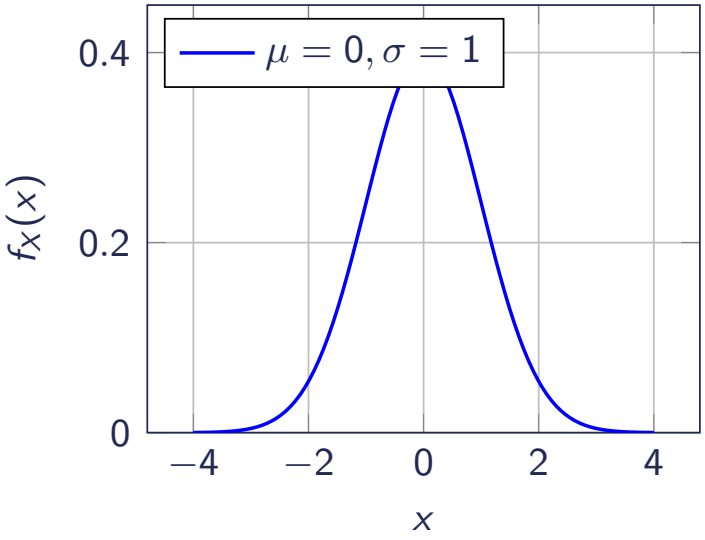
Properties:

- Symmetric around μ
- 68% within $\mu \pm \sigma$
- 95% within $\mu \pm 2\sigma$

Why is it important?

Parameters:

- μ: mean (center)
- σ^2: variance (spread)



Caution!

Many phenomena in the social systems **DO NOT** follow Normal distribution. Check out **Powerlaw** distribution.

Joint Probability Distribution

Question: How do we represent probability distributions when we have *multiple* random variables?

Definition

The **joint PDF** $f_{X,Y}(x, y)$ gives the probability of observing both $X = x$ AND $Y = y$:

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

Example: Roll a die (X) and flip an unfair coin (Y)

$Y \backslash X$	$X = 1$	$X = 2$	$X = 3$	$X = 4$	$X = 5$	$X = 6$
$Y = H$	$p(1, H)$	$p(2, H)$	$p(3, H)$	$p(4, H)$	$p(5, H)$	$p(6, H)$
$Y = T$	$p(1, T)$	$p(2, T)$	$p(3, T)$	$p(4, T)$	$p(5, T)$	$p(6, T)$

Each cell: probability of that specific combination of outcomes

Marginal Distributions

Question: Given joint distribution, how do we get individual (**marginal**) distributions?

Answer: Sum (or integrate) over the other variable

Marginalization

Discrete: $$P(X = x) = \sum_y P(X = x, Y = y)$$

Continuous: $$f_X(x) = \int f_{X,Y}(x, y)\, dy$$

Example: Die + Coin

$$P(X = 1) = P(X = 1, Y = H) + P(X = 1, Y = T)$$

$$P(Y = H) = \sum_{i=1}^{6} P(X = i, Y = H)$$

Independence

Definition

Two random variables X and Y are **independent** if:

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y) \quad \text{for all } x, y$$

Intuition: Knowing Y tells you nothing about X

Notation: $X \perp Y$ (read: "X independent of Y")

Independent example:

- Roll die, flip coin
- $P(\text{die} = 3, \text{coin} = H) = P(\text{die} = 3) \cdot P(\text{coin} = H)$

Dependent example:

- Height and weight of people
- $P(H = h, W = w) \neq P(H = h) \cdot P(W = w)$

Random Variable Calculus

Question: What happens when we perform operations on RVs?

Examples:

- $Z = 2X + 3$ (linear trans.)
- $Z = X + Y$ (sum)
- $Z = X - Y$ (difference)
- $Z = XY$ (product)

Answer: It depends on:

- The distributions of X and Y
- Whether they are **independent**

Key Results for Independent Gaussians

If $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ are independent:

$$aX + b \sim \mathcal{N}(a\mu_X + b, a^2\sigma_X^2)$$
$$X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$
$$X - Y \sim \mathcal{N}(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

Exercise: Linear Transformations

Given: $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ are independent.

Find the distribution of Z when:

1. $Z = 2X + 3$
2. $Z = X + Y$
3. $Z = X - Y$
4. $Z = \frac{X+Y}{2}$ (average)

Hint: Use the formulas from the previous slide!

Exercise: Solutions

Given: $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$, $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ independent

1. $Z = 2X + 3$

$$Z \sim \mathcal{N}(2\mu_X + 3,\, 4\sigma_X^2)$$

2. $Z = X + Y$

$$Z \sim \mathcal{N}(\mu_X + \mu_Y,\, \sigma_X^2 + \sigma_Y^2)$$

3. $Z = X - Y$

$$Z \sim \mathcal{N}(\mu_X - \mu_Y,\, \sigma_X^2 + \sigma_Y^2)$$

4. $Z = \frac{X+Y}{2}$

$$Z \sim \mathcal{N}\left(\frac{\mu_X + \mu_Y}{2},\, \frac{\sigma_X^2 + \sigma_Y^2}{4}\right)$$

Variances always add (for independent variables)!

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Conditional Probability

Motivation: Often we gain information about one variable that affects another, how do we formalize it?

Definition

The **conditional probability** of $X = x$ given $Y = y$ is:

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Intuition: Among all cases where $Y = y$, what fraction has $X = x$?

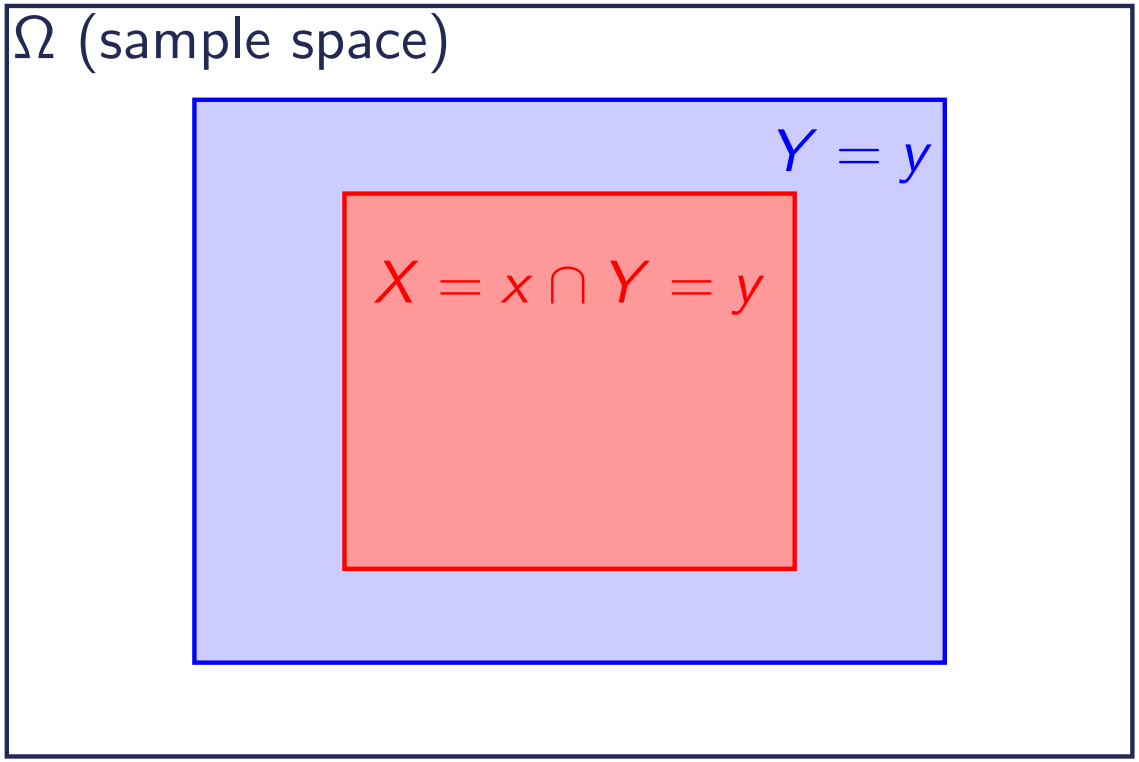
Key property: $P(X = x, Y = y) = P(Y = y) \cdot P(X = x \mid Y = y)$

Symmetry: $P(X = x, Y = y) = P(X = x) \cdot P(Y = y \mid X = x)$

Therefore:

$$P(Y = y) \cdot P(X = x \mid Y = y) = P(X = x) \cdot P(Y = y \mid X = x)$$

Visualizing Conditional Probability



Conditional probability = restricting attention to the blue region, asking what fraction is red

$$P(X = x \mid Y = y) = \frac{\text{Red area}}{\text{Blue area}} = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Deriving Bayes' Law

Starting point: Two expressions for joint probability

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y \mid X = x)$$
$$P(X = x, Y = y) = P(Y = y) \cdot P(X = x \mid Y = y)$$

Set them equal:

$$P(X = x) \cdot P(Y = y \mid X = x) = P(Y = y) \cdot P(X = x \mid Y = y)$$

Solve for $P(X = x \mid Y = y)$:

Bayes' Law

$$P(X = x \mid Y = y) = \frac{P(Y = y \mid X = x) \cdot P(X = x)}{P(Y = y)}$$

Bayes' Law: Bayesian Terminology

$$P(\theta \mid D) = \frac{P(D \mid \theta) \cdot P(\theta)}{P(D)}$$

- θ: **Parameter** or hypothesis (what we want to learn about)
- D: **Data** or evidence (what we observe)

The Four Components

Prior	$P(\theta)$: Belief about θ *before* seeing data
Likelihood	$P(D \mid \theta)$: Probability of data given parameter
Posterior	$P(\theta \mid D)$: Updated belief about θ *after* seeing data
Evidence	$P(D)$: Normalizing constant (often hard to compute)

Proportional form (often used):

$$P(\theta \mid D) \propto P(D \mid \theta) \cdot P(\theta)$$

Law of Total Probability

Problem: How do we compute $P(D)$?

Law of Total Probability

If events $\theta_1, \theta_2, \ldots, \theta_n$ partition the sample space:

$$P(D) = \sum_{i=1}^{n} P(D \mid \theta_i) \cdot P(\theta_i) \xrightarrow{\textit{continuous } \theta} \int P(D \mid \theta) \cdot P(\theta)\, d\theta$$

Intuition: Average the likelihood over all possible parameter values, weighted by their prior probabilities

Complete Bayes' Law:

$$P(\theta \mid D) = \frac{P(D \mid \theta) \cdot P(\theta)}{\sum_i P(D \mid \theta_i) \cdot P(\theta_i)}$$

Example 1: Medical Testing

Scenario: A diagnostic test for a rare cancer

Given information:

- Prevalence: $P(\text{cancer}) = 0.01$ (1%)
- True positive rate: $P(\text{positive} \mid \text{cancer}) = 0.90$ (90%)
- False positive rate: $P(\text{positive} \mid \text{no cancer}) = 0.05$ (5%)

Question: A person tests positive. What is the probability they actually have cancer?

Medical Testing: Setup

Define random variables:

$$X = \begin{cases} x_1 & \text{person has cancer} \\ x_2 & \text{person have no cancer} \end{cases} \qquad Y = \begin{cases} y_1 & \text{test result is positive} \\ y_2 & \text{test result is negative} \end{cases}$$

Known probabilities:

$$P(X = x_1) = 0.01 \qquad \text{(prevalence)}$$
$$P(X = x_2) = 0.99 \qquad \text{(complement)}$$
$$P(Y = y_1 \mid X = x_1) = 0.90 \qquad \text{(true positive rate)}$$
$$P(Y = y_1 \mid X = x_2) = 0.05 \qquad \text{(false positive rate)}$$

Goal: Find $P(X = x_1 \mid Y = y_1)$ (probability of cancer given positive test)

Medical Testing: Solution

Apply Bayes' Law:

$$P(X = x_1 \mid Y = y_1) = \frac{P(Y = y_1 \mid X = x_1) \cdot P(X = x_1)}{P(Y = y_1)}$$

Step 1: Compute $P(Y = y_1)$ using Law of Total Probability

$$\begin{aligned} P(Y = y_1) &= P(Y = y_1 \mid X = x_1) \cdot P(X = x_1) \\ &\quad + P(Y = y_1 \mid X = x_2) \cdot P(X = x_2) \\ &= (0.90)(0.01) + (0.05)(0.99) \\ &= 0.009 + 0.0495 = 0.0585 \end{aligned}$$

Step 2: Substitute into Bayes' Law

$$P(X = x_1 \mid Y = y_1) = \frac{(0.90)(0.01)}{0.0585} = \frac{0.009}{0.0585} \approx 0.154$$

Medical Testing: Interpretation

Result: $\boxed{P(\text{cancer} \mid \text{positive test}) \approx 15.4\%}$

Surprising! Despite a 90% accurate test, positive result only gives 15% chance of cancer.

Why so low?

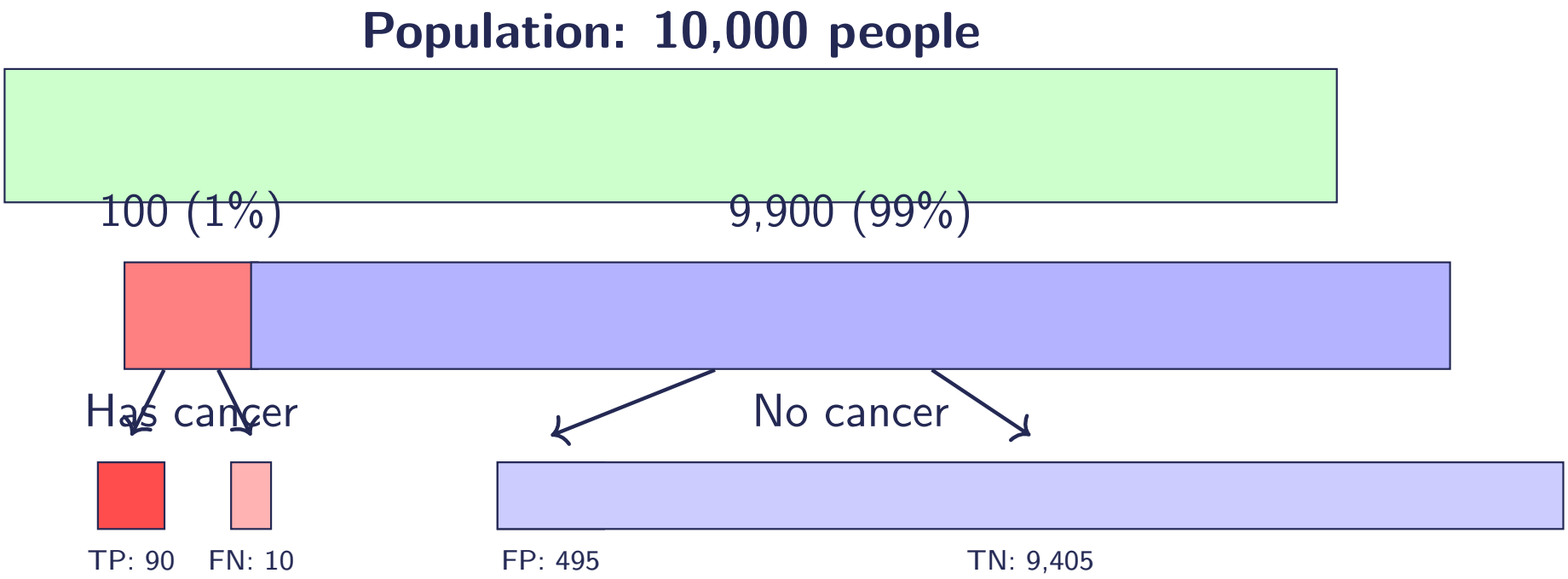
1. Disease is **rare** (1% prevalence)
2. Many more healthy people than sick people
3. 5% of healthy people test positive (false positives)
4. False positives from large healthy population outnumber true positives from small sick population

Key Insight

The **prior probability** (prevalence) heavily influences the posterior. Bayes' Law properly balances:

Evidence from test (likelihood) & Base rate of disease (prior)

Visualizing the Medical Test Problem



Positive tests: $90 + 495 = 585$

True positives: 90

$$P(\text{cancer} \mid +) = \frac{90}{585} \approx 15.4\%$$

Sequential Updating

Beautiful property of Bayes' Law: Can update sequentially!

Sequential Bayes

After observing data D_1:

$$P(\theta \mid D_1) = \frac{P(D_1 \mid \theta) \cdot P(\theta)}{P(D_1)}$$

After observing more data D_2:

$$P(\theta \mid D_1, D_2) = \frac{P(D_2 \mid \theta) \cdot P(\theta \mid D_1)}{P(D_2 \mid D_1)}$$

Key: Today's posterior = tomorrow's prior!

Medical test: What would be the probability of the person actually having the disease if the result of the second test is also positive?

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What We Learned Today

- ✓ **Random variables:** Quantifying uncertainty
- ✓ **Probability distributions:** PDF, CDF, joint distributions
- ✓ **Independence:** Variables tell us nothing about each other
- ✓ **Conditional probability:** Restricting to subsets of sample space
- ✓ **Bayes' Law:** Derived from conditional probabilities
- ✓ **Prior, Likelihood, Posterior:** The Bayesian update cycle
- ✓ **Python implementation:** Discrete Bayes updates

Key Conceptual Insight

Bayesian inference is a **principled way to learn from data** by combining:

- Prior beliefs (what we think before observing)
- Evidence (likelihood of data under each hypothesis)
- Posterior beliefs (updated knowledge after observing)

Next Session

Bayesian Inference: Prior, Likelihood, Posterior

Topics:

- Prior specification philosophy
- Likelihood: The generative model
- Posterior: Combining prior and likelihood
- Conjugate families (analytical solutions)
- More PyMC

Building on today:

- Today: Discrete hypotheses, manual calculation
- Next session: Continuous parameters, computational methods

Homework

Before next session:

1. **Practice Bayes' Law:**
   - Work through medical testing example with different numbers
   - Exercise: "rare disease with 99% accurate test and 0.1% prevalence"
2. **Python exercises:**
   - Implement sequential Bayes update
   - Visualize convergence as data accumulates
   - Experiment with different priors
3. **Reading:**
   - Gill Ch. 2 (Bayes' Theorem)
4. **Play around PyMC**, e.g, how to define data, priors, likelihood etc.

Questions?

Next: Bayesian Reasoning